

FCC RF EXPOSURE REPORT

FCC ID: HLZT7

Report No. : BTL-FCCP-5-2311H013
Equipment : Connect T7 Wi-Fi 7 Mesh Router
Model Name : T7
Brand Name : Predator
Applicant : Acer Incorporated
Address : 8F, 88, Sec. 1, Xintai 5th Rd., Xizhi, New Taipei City 221, Taiwan, R.O.C.

Radio Function : WLAN 2.4 GHz, RLAN 5 GHz (U-NII 1, U-NII 2A, U-NII 2C, U-NII 3),
U-NII 6 GHz (U-NII 5, U-NII 6, U-NII 7, U-NII 8)

FCC Rule Part(s) : FCC Guidelines for Human Exposure IEEE C95.1 & FCC Part 2.1091
FCC Title 47 Part 2.1091

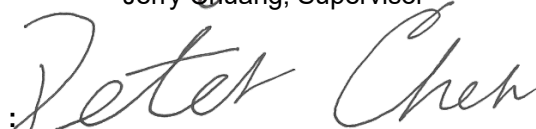
Date of Receipt : 2023/11/9
Date of Test : 2023/12/4 ~ 2024/3/14
Issued Date : 2024/6/12

The above equipment has been tested and found in compliance with the requirement of the above standards by BTL Inc.

Prepared by


Jerry Chuang, Supervisor

Approved by


Peter Chen, Manager



BTL Inc.

No.18, Ln. 171, Sec. 2, Jiuzong Rd., Neihu Dist., Taipei City 114, Taiwan

Tel: +886-2-2657-3299 Fax: +886-2-2657-3331 Web: www.newbtl.com Service mail: btl_qa@newbtl.com

REPORT ISSUED HISTORY

Report No.	Version	Description	Issued Date	Note
BTL-FCCP-5-2311H013	R00	Original Report.	2024/5/20	Invalid
BTL-FCCP-5-2311H013	R01	Revised report to address comments.	2024/6/12	Valid

1. TEST FACILIT

The test locations stated below are under the TAF Accreditation Number 0659.

The test location(s) used to collect the test data in this report are:

No. 68-1, Ln. 169, Sec. 2, Datong Rd., Xizhi Dist., New Taipei City 221, Taiwan
(FCC DN: TW0659)

☐ C05 ☐ CB08 ☐ CB11 ☐ SR10 ☒ SR11

2. MPE CALCULATION METHOD

Calculation Method of RF Safety Distance:

$$S = \frac{PG}{4\pi^2} = \frac{EIRP}{4\pi^2}$$

where:

S = power density

P = power input to the antenna

G = power gain of the antenna in the direction of interest relative to an isotropic radiator

R = distance to the center of radiation of the antenna

3. ANTENNA SPECIFICATION

For 2.4GHz:

Ant.	Brand	Part number	Type	Connector	Gain (dBi)
1		SH23227IB65-1	PIFA	I-PEX	1.43
2		SH23227IB65-2	PIFA	I-PEX	0.84

NOTE:

- The EUT incorporates a CDD function. Physically, the EUT provides two completed transmitters and receivers (2T2R).
- For Output Power
For $N_{ANT} = 2 < 5$,
Direction gain = $G_{ANT} + 0 = 1.43 + 0 = 1.43$ dBi.
The Direction gain is less than 6 dBi, so output power limits will not be reduced.
- For Beamforming:
Beamforming Gain: 3dB
so Directional gain = $1.43 + 3 = 4.43$ dBi, so output power limits will not be reduced.

For 5GHz:

Ant.	Brand	Part number	Type	Connector	Gain (dBi)
1		SH23227IB65-1	PIFA	I-PEX	5.38
2		SH23227IB65-2	PIFA	I-PEX	4.06

Note:

- The EUT incorporates a CDD function. Physically, the EUT provides two completed transmitters and receivers (2T2R).
- For Output Power
For No Beamforming, $N_{ANT} = 2 < 5$; so Directional gain = 5.38.
The Direction gain is less than 6 dBi, so output power limits will not be reduced.
For Beamforming, Beamforming Gain: 3 dBi, so Directional gain = $5.38 + 3 = 8.38$ dBi.
To UNII-1, UNII-3, the reduced output power limits (dBm) = $30 - (8.38 - 6) = 27.62$.
To UNII-2A, UNII-2C, the reduced output power limits (dBm) = $24 - (8.38 - 6) = 21.62$
- For Power Spectral Density
Directional Gain = $10 \log [(10^{G1/20} + 10^{G2/20} + \dots + 10^{Gn/20})^2 / N_{ANT}] = 7.76$ dBi > 6dBi.
To UNII-1, the reduced power spectral density limits (dBm/MHz) = $17 - (7.76 - 6) = 15.24$.
To UNII-2A, UNII-2C, the reduced power spectral density limits (dBm/MHz) = $11 - (7.76 - 6) = 9.24$.
To UNII-3, the reduced power spectral density limits (dBm/500 kHz) = $30 - (7.76 - 6) = 28.24$.

For WIFI 6E:

Ant.	Brand	Part number	Type	Connector	Frequency Range (MHz)	Gain (dBi)
1		SH23227IB65-1	PIFA	I-PEX	5950-7120	3.23
2		SH23227IB65-2	PIFA	I-PEX	5950-7120	3.61

Note:

- CDD Directional Gain = Antenna Gain + Array Gain, Array Gain = 0 dB for $N_{ANT} \leq 4$;
CDD Directional Gain = $3.61 + 0 = 3.61$
Beamforming Directional Gain = Antenna Gain + $10 \cdot \log (N_{ANT}/N_{SS})$.
Beamforming Directional Gain = $3.61 + 3 = 6.61$

4. TABLE FOR ANTENNA CONFIGURATION

For 2.4GHz:

For No Beamforming:

TX Mode \ Operating Mode	2TX
IEEE 802.11b	V (Ant. 1+Ant. 2)
IEEE 802.11g	V (Ant. 1+Ant. 2)
IEEE 802.11n (HT20)	V (Ant. 1+Ant. 2)
IEEE 802.11n (HT40)	V (Ant. 1+Ant. 2)
IEEE 802.11ax (HE20)	V (Ant. 1+Ant. 2)
IEEE 802.11ax (HE40)	V (Ant. 1+Ant. 2)
IEEE 802.11be (EHT20)	V (Ant. 1+Ant. 2)
IEEE 802.11be (EHT40)	V (Ant. 1+Ant. 2)

For Beamforming:

TX Mode \ Operating Mode	2TX
IEEE 802.11n (HT20)	V (Ant. 1+Ant. 2)
IEEE 802.11n (HT40)	V (Ant. 1+Ant. 2)
IEEE 802.11ax (HE20)	V (Ant. 1+Ant. 2)
IEEE 802.11ax (HE40)	V (Ant. 1+Ant. 2)
IEEE 802.11be (EHT20)	V (Ant. 1+Ant. 2)
IEEE 802.11be (EHT40)	V (Ant. 1+Ant. 2)

For 5GHz:

For No Beamforming

TX Mode \ Operating Mode	2TX
IEEE 802.11a	V (Ant. 1+Ant. 2)
IEEE 802.11n (HT20)	V (Ant. 1+Ant. 2)
IEEE 802.11n (HT40)	V (Ant. 1+Ant. 2)
IEEE 802.11ac (VHT80)	V (Ant. 1+Ant. 2)
IEEE 802.11ac (VHT160)	V (Ant. 1+Ant. 2)
IEEE 802.11ax (HE20)	V (Ant. 1+Ant. 2)
IEEE 802.11ax (HE40)	V (Ant. 1+Ant. 2)
IEEE 802.11ax (HE80)	V (Ant. 1+Ant. 2)
IEEE 802.11ax (HE160)	V (Ant. 1+Ant. 2)
IEEE 802.11be (EHT20)	V (Ant. 1+Ant. 2)
IEEE 802.11be (EHT40)	V (Ant. 1+Ant. 2)
IEEE 802.11be (EHT80)	V (Ant. 1+Ant. 2)
IEEE 802.11be (EHT160)	V (Ant. 1+Ant. 2)

For Beamforming

TX Mode	Operating Mode	2TX
IEEE 802.11n (HT20)		V (Ant. 1+Ant. 2)
IEEE 802.11n (HT40)		V (Ant. 1+Ant. 2)
IEEE 802.11ac (VHT80)		V (Ant. 1+Ant. 2)
IEEE 802.11ac (VHT160)		V (Ant. 1+Ant. 2)
IEEE 802.11ax (HE20)		V (Ant. 1+Ant. 2)
IEEE 802.11ax (HE40)		V (Ant. 1+Ant. 2)
IEEE 802.11ax (HE80)		V (Ant. 1+Ant. 2)
IEEE 802.11ax (HE160)		V (Ant. 1+Ant. 2)
IEEE 802.11be (EHT20)		V (Ant. 1+Ant. 2)
IEEE 802.11be (EHT40)		V (Ant. 1+Ant. 2)
IEEE 802.11be (EHT80)		V (Ant. 1+Ant. 2)
IEEE 802.11be (EHT160)		V (Ant. 1+Ant. 2)

For WIFI 6E:

TX Mode	Operating Mode	2TX
IEEE 802.11ax (HE20)		V (Ant. 1 + Ant. 2)
IEEE 802.11ax (HE40)		V (Ant. 1 + Ant. 2)
IEEE 802.11ax (HE80)		V (Ant. 1 + Ant. 2)
IEEE 802.11ax (HE160)		V (Ant. 1 + Ant. 2)
IEEE 802.11be (EHT20)		V (Ant. 1 + Ant. 2)
IEEE 802.11be (EHT40)		V (Ant. 1 + Ant. 2)
IEEE 802.11be (EHT80)		V (Ant. 1 + Ant. 2)
IEEE 802.11be (EHT160)		V (Ant. 1 + Ant. 2)
IEEE 802.11be (EHT320)		V (Ant. 1 + Ant. 2)

5. CALCULATED RESULT

For 2.4GHz_Non Beamforming:

Direction gain (dBi)	Direction gain (numeric)	Max. Output Power (dBm)	Max. Output Power (mW)	Power Density (S) (mW/cm ²)	Limit of Power Density (S) (mW/cm ²)	Test Result
1.43	1.3900	25.55	358.9219	0.0993	1	Complies

For 2.4GHz_Beamforming:

Direction gain (dBi)	Direction gain (numeric)	Max. Output Power (dBm)	Max. Output Power (mW)	Power Density (S) (mW/cm ²)	Limit of Power Density (S) (mW/cm ²)	Test Result
4.43	2.7733	24.98	314.7748	0.1738	1	Complies

For 5GHz_Non Beamforming:

Direction gain (dBi)	Direction gain (numeric)	Max. Output Power (dBm)	Max. Output Power (mW)	Power Density (S) (mW/cm ²)	Limit of Power Density (S) (mW/cm ²)	Test Result
5.38	3.4514	27.32	539.5106	0.3706	1	Complies

For 5GHz_Beamforming:

Direction gain (dBi)	Direction gain (numeric)	Max. Output Power (dBm)	Max. Output Power (mW)	Power Density (S) (mW/cm ²)	Limit of Power Density (S) (mW/cm ²)	Test Result
8.38	6.8865	26.44	440.5549	0.6039	1	Complies

For WIFI 6E_Non Beamforming:

Max. e.i.r.p. (dBm)	Max. e.i.r.p. (mW)	Power Density (S) (mW/cm ²)	Limit of Power Density (S) (mW/cm ²)	Test Result
26.12	409.2607	0.0815	1	Complies

For WIFI 6E_Beamforming:

Max. e.i.r.p. (dBm)	Max. e.i.r.p. (mW)	Power Density (S) (mW/cm ²)	Limit of Power Density (S) (mW/cm ²)	Test Result
28.67	736.2071	0.1465	1	Complies

For the max simultaneous transmission MPE:

Ratio			Total	Limit of Ratio	Test Result
2.4GHz	5GHz	6E			
0.1738	0.6039	0.14654	0.9242	1	Complies

Note: The calculated distance is 20 cm.

End of Test Report